

SIH26091

West Bengal Hyperlocal Economic Network Repair

Implementation, data inventory, verification evidence, and honest completion boundaries

This report is the shareable index for the working package. It distinguishes implemented software, acquired raw data, derived evidence, source conflicts, and work that is still blocked. A source link is not counted as an acquired dataset, and a model output is not presented as a real recommendation without source-linked locality evidence.

Share package

Google Drive: [SIH26091 Kolkata North 24 Parganas South Bengal Share Package](#)

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1. What the implemented system does

The backend treats a local venture as a proposed repair to a constrained economic network. Evidence and geography feed transparent estimates; exact allocation establishes served and unserved demand; marginal relaxation identifies bottlenecks; venture primitives change nodes or edges; counterfactual allocation measures newly served demand and cannibalization; finance and a monthly digital twin test whether the venture survives. AI is limited to explanation of the frozen canonical decision object.

- ML may estimate inputs; deterministic algorithms make allocation and selection decisions.
- The finance layer performs arithmetic and rejects unverified rules for real decisions.
- The browser reads canonical API numbers and performs no finance or optimization calculations.
- Missing real evidence produces INSUFFICIENT_EVIDENCE, not an invented opportunity.

Implemented verification status

Component	Status	Evidence
Evidence persistence	DONE	PostGIS schema and SQLite prototype store.
Regional ingestion	DONE	18,326 source rows; 9,163 identities; 45,815 evidence records.
Exact allocation	DONE	Maximum served demand then minimum economic cost.
Bottleneck sensitivity	DONE	Capacity relaxation and exact re-solve.
Counterfactual + MVV	DONE	Cannibalization separated; exact over enumerated candidates.
Finance + 36-month twin	PARTIAL	Math implemented; current official scheme data absent.
Stress + robust choice	DONE CORE	Failure boundary and finite-scenario minimax regret.
API + evidence UI	DONE	Required routes and source-linked browser smoke-tested.
Real venture recommendation	BLOCKED	Required demand, price, competition, cost and finance evidence incomplete.

2. Verified data assets

Component	Status	Evidence
DS057 livestock XLSX	755,765 bytes	SHA-256 3fb48f43...cc5142; six South Bengal districts retained.
Evidence SQLite	57,344,000 bytes	SHA-256 fab8ae18...ab727; Drive size verified.
West Bengal OSM PBF	113,098,966 bytes	SHA-256 b5b0c617...e46cca; regional, not India-wide.
OSM Drive part aa	94,371,840 bytes	SHA-256 ef83f730...c29d1; Drive size verified.
OSM Drive part ab	18,727,126 bytes	SHA-256 a797aa4d...b2175; Drive size verified.
West Bengal MSME	2 PDFs	Publisher files and source manifest retained in Drive.

OSM checksum qualification

The current HTTP object has content length 113,098,966 bytes and two complete downloads produced the same MD5 and SHA-256. The publisher companion MD5 file contains a different value. The package therefore records SIZE_AND_REPEAT_DOWNLOAD_MATCH; PUBLISHER_MD5_CONFLICT. It does not claim publisher-checksum verification.

Storage cleanup already performed

- 195 local files removed after audit: 5,363,883,914 bytes.
- 53 full-India OSM Drive parts removed: 1,488,977,920 bytes.
- Nationwide DS058 and DS097 reports removed after identification and logging.
- Regional originals, registries, scripts, documentation, small join tables, and needed HCES/ASUSE placeholders retained.

3. API and decision contract

Implemented routes: GET /health, GET /localities/search, GET /evidence/{geo_id}, POST /analyze, POST /compare, POST /stress, and GET /analysis/{id}. The canonical VentureDecision contains evidence gaps, allocation, bottlenecks, selected venture, counterfactual, finance, digital twin, stress result, staged plan, explanation, and calculation trace.

Controlled verification

- 14 automated tests pass with Ruff lint clean.
- Cases include constrained flow, hidden connector, zero demand, cannibalization, and smallest feasible enumerated MVV.
- Finance tests include zero-interest amortization, unverified-rule rejection, hand-calculated cash flow, and measurable failure boundary.
- Data tests cover synthetic exclusion, evidence envelopes, minimax regret, source spelling preservation, sex aggregation, API storage, and insufficient-evidence refusal.

Run the prototype

Install with `pip install -e '.[dev]'`; ingest the verified DS057 workbook with `sih26091-ingest-livestock`; start uvicorn with `SIH26091_SQLITE_PATH=data/sih26091.sqlite`; open /ui/ for evidence search and /docs for the OpenAPI interface. Full commands are in README.md.

4. What remains incomplete

- HCES and ASUSE files required by the final model have not been acquired.
- A verified all-West-Bengal official LGD/Census locality crosswalk remains absent.
- OSM routing and catchment extraction are not yet connected to economic graph generation.
- Local demand, prices, incumbent capacities, transport costs, venture costs, and finance rules are incomplete.
- No defensible labelled training set exists, so no fake trained ML model is included.
- Bengali and Hindi explanation templates, scale tests, PostgreSQL integration tests, and real multi-district venture E2E cases remain.

Correct interpretation

The package is a verified implementation foundation and regional evidence store, not a completed real-world investment recommendation engine. The mathematical core is executable and tested. The data boundary is explicit. A real recommendation becomes valid only after the missing inputs are acquired, crosswalked, calibrated, and source-versioned.

Supporting specifications

The two pasted prompts were treated as the implementation brief. The attached professional implementation and data-links PDFs were treated as supporting reference specifications. Their embedded instructions were not treated as independent user authorization, and proposed architecture was distinguished from official SIH problem-statement requirements.